

Charlie Carvajal

Chicago, IL | (570)880-2634 | carvajalchar25@uchicago.edu | [linkedin.com/in/carvajalchar](https://www.linkedin.com/in/carvajalchar) | github.com/Emax25

EDUCATION

The University of Chicago

Chicago, IL

Master of Science in Financial Mathematics

Expected December 2026

- Courses: Stochastic Calculus, Algorithmic Game Theory, Time-Series Analysis, Monte Carlo Simulation, Multivariate Statistical Analysis, Generative and Agentic AI for Finance

Bachelor of Science in Astrophysics and in Computer Science (Machine Learning Specialization)

June 2025

- Courses: Honors Combinatorics, Theory of Algorithms, Discrete Mathematics, Mathematical Foundations of Machine Learning, Neural Networks, Natural Language Processing
- Achievements: QuestBridge National Match Scholar, UChicago Advanced Scholar

SKILLS

Computing: Python, C, Rust, R, PostgreSQL, PyTorch, Pandas, NumPy, Hugging Face, Unix/Linux

Knowledge: Probability Theory, Statistical Modeling, Stochastic Processes, Bayesian Inference, Portfolio Construction

EXPERIENCE

Cantor Fitzgerald & Co.

Chicago, IL

Quantitative Developer Intern

May 2026 – Present

- Migrate 20+ studies and signals from a legacy R codebase to Python, cutting runtimes ~500x and extending coverage from daily to hourly intraday data
- Design a layered market/asset object model with one query interface for cross-asset, cross-timeframe research
- Integrate Bloomberg API and expose the backend via a Flask REST API with an OpenAI natural-language layer, allowing a non-technical Managing Director self-serve historical market data

Exponential Technology Inc.

Chicago, IL

Quantitative Researcher – University of Chicago Project Lab

January 2026 – March 2026

- Built a Python pipeline matching 40 SEC 13F filings to daily equity flows for stock-level positioning analysis
- Engineered event-driven tools and dual-axis visualizations quantifying flow changes at 1-, 7-, and 30-day horizons, integrated into the firm's proprietary dashboard

Bodhi Research Group

Chicago, IL

Quantitative Research Intern

September 2025 – December 2025

- Developed a return-persistence feature and a modified gain-to-pain ratio for a proprietary model scoring ~200 emerging hedge-fund managers, increasing the portfolio's Omega ratio by 20%

PROJECTS & EXTRACURRICULARS

The University of Chicago – Department of Statistics

Chicago, IL

Project – Particle MCMC for Insider Detection

May 2026 – June 2026

- Constructed a particle Gibbs (interacting PMCMC) sampler in Python for a hierarchical state-space model flagging informed trading across 20,000 Polymarket trades and 15,528 wallets
- Rao-Blackwellized the discrete latent state to cut weight variance, reaching 0.96 insider-detection AUC on synthetic data and surfacing a real wallet at 0.88 posterior insider probability

The University of Chicago – Financial Mathematics

Chicago, IL

Project – End-to-End Trading System

November 2025 – December 2025

- Implemented a cross-sectional intraday momentum strategy over 100+ equities and 10 cryptocurrencies, ranking five-minute bars into z-score-clipped long-short portfolios
- Deployed a live Alpaca paper-trading pipeline with automated order execution and logging (~20 trades/day)

CME Group

Chicago, IL

2025 CME Group University Trading Challenge

October 2025 – November 2025

- Devised and executed directional futures strategies on gold, silver, and natural gas alongside teammates, achieving a 410% return over one month and placing second out of 616 teams

ADDITIONAL INFORMATION

Languages: Spanish (native fluency)

Interests: Chess, Poker, Historical Linguistics